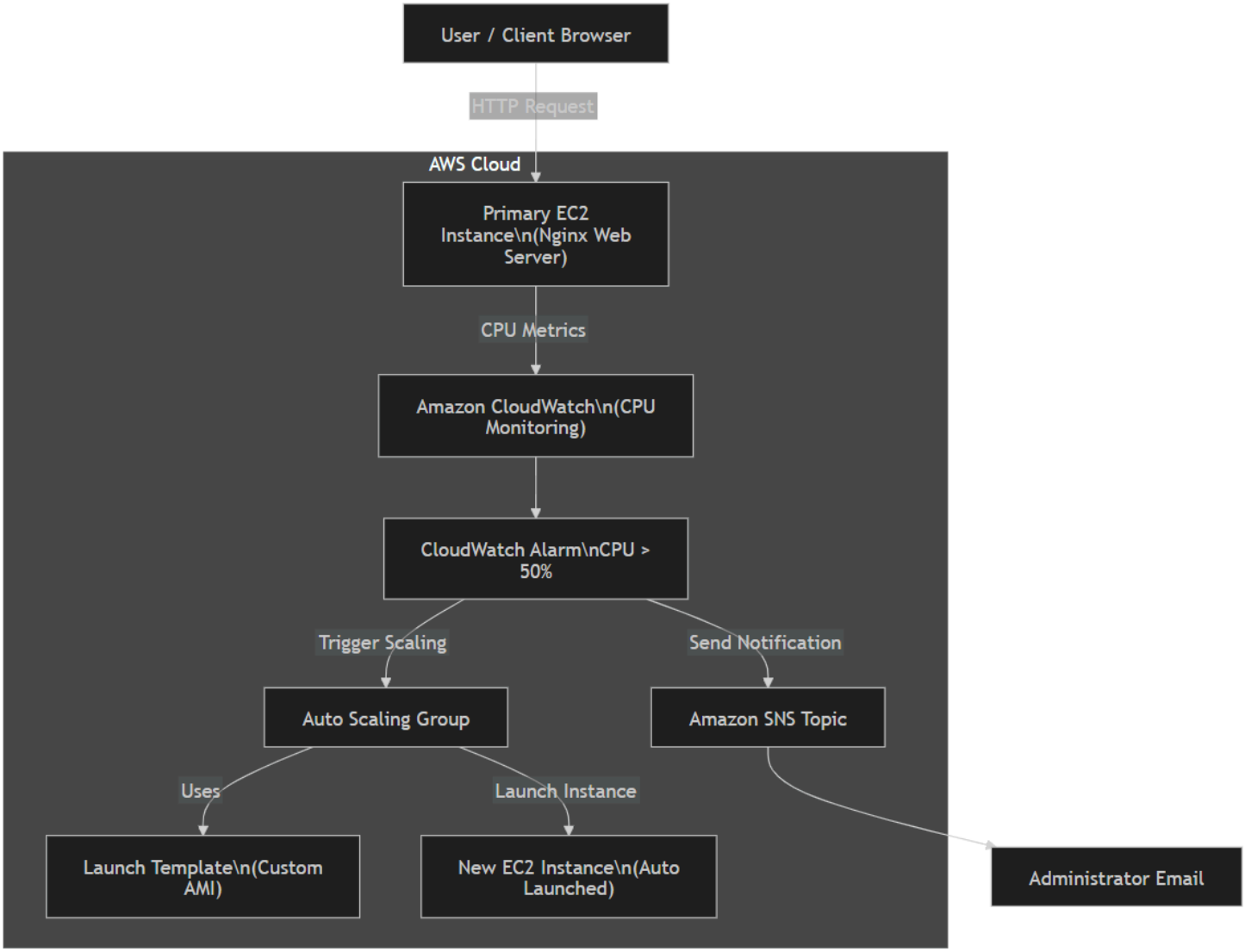
SmartScale: Intelligent EC2 Auto Scaling and Alerting Platform

Project Overview

SmartScale is an enterprise-style AWS automation project that demonstrates intelligent infrastructure scaling based on real-time CPU utilization. The system continuously monitors EC2 performance and automatically launches new EC2 instances when CPU usage exceeds the defined threshold, ensuring high availability and performance stability.
In addition, real-time email notifications are sent to administrators during scaling events using Amazon SNS.
This project simulates real-world production architecture and follows AWS best practices for monitoring, scaling, and alerting.



Problem Statement

Traditional server-based systems suffer from:

- Fixed infrastructure capacity
- Manual scaling delays
- Risk of downtime during traffic spikes
- Inefficient resource utilization

SmartScale solves these problems through automated, event-driven scaling.

Project Objectives

- Automate EC2 scaling based on CPU utilization

- Maintain identical configuration across instances
- Implement proactive alerting via email
- Achieve high availability and fault tolerance
- Reduce manual operational overhead

AWS Services Used

- Amazon EC2
- Amazon Machine Image (AMI)
- Launch Template
- Auto Scaling Group (ASG)
- Amazon CloudWatch
- Amazon SNS
- Amazon VPC
- Security Groups
- Nginx

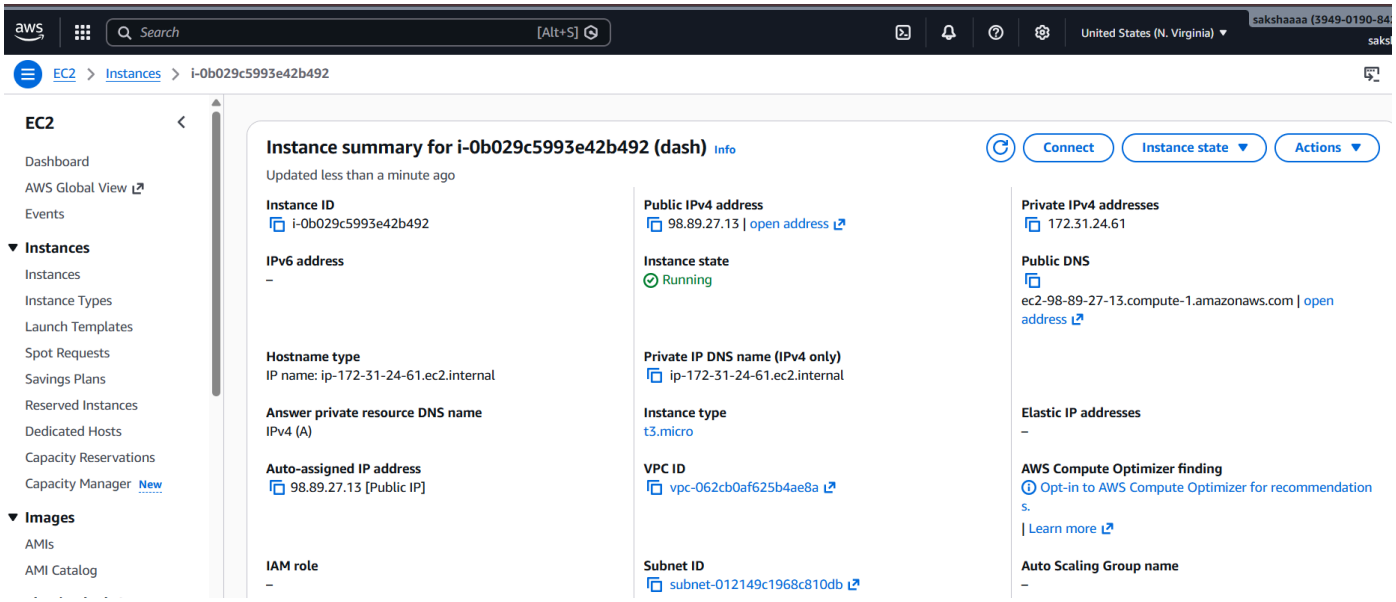
Architecture Flow

1. User accesses the application hosted on EC2 via Nginx
2. CloudWatch continuously monitors CPU utilization
3. When CPU exceeds 50%, CloudWatch alarm triggers
4. Auto Scaling policy executes
5. Auto Scaling Group launches a new EC2 instance
6. Launch Template uses custom AMI for identical configuration
7. New instance becomes healthy and serves traffic
8. SNS sends real-time email notification

Step-by-Step Implementation

Step 1 : EC2 Setup and Web Server Configuration

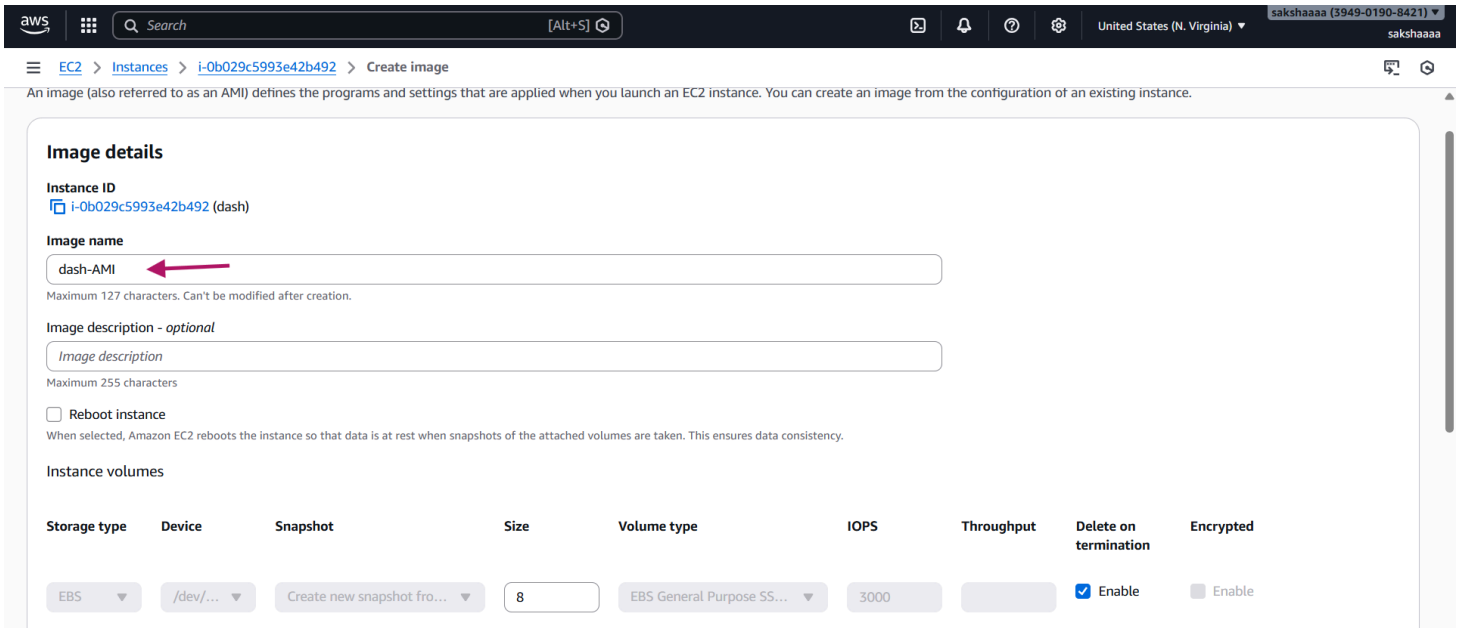
- Launch EC2 instance (Amazon Linux)



- open linux terminal : Enter below command
- ssh -i ./downloads/key.pem ec2-user@ip
- Install and configure Nginx
- sudo yum update -y
- sudo yum install nginx -y
- sudo systemctl start nginx
- sudo systemctl enable nginx
- Verify application via public IP

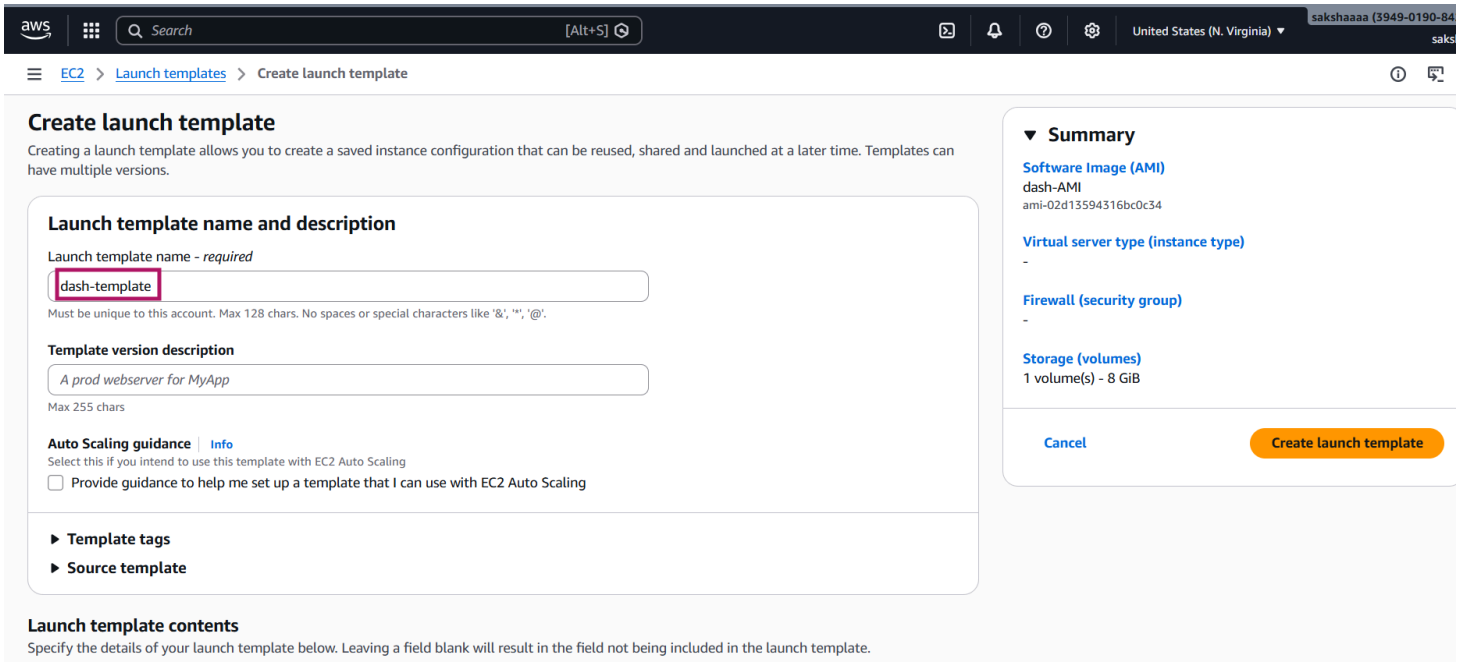
Step 2 : Create Custom AMI

- Select configured EC2 instance
- Create AMI to preserve configuration
- Wait until AMI status is available



Step 3 : Configure Launch Template

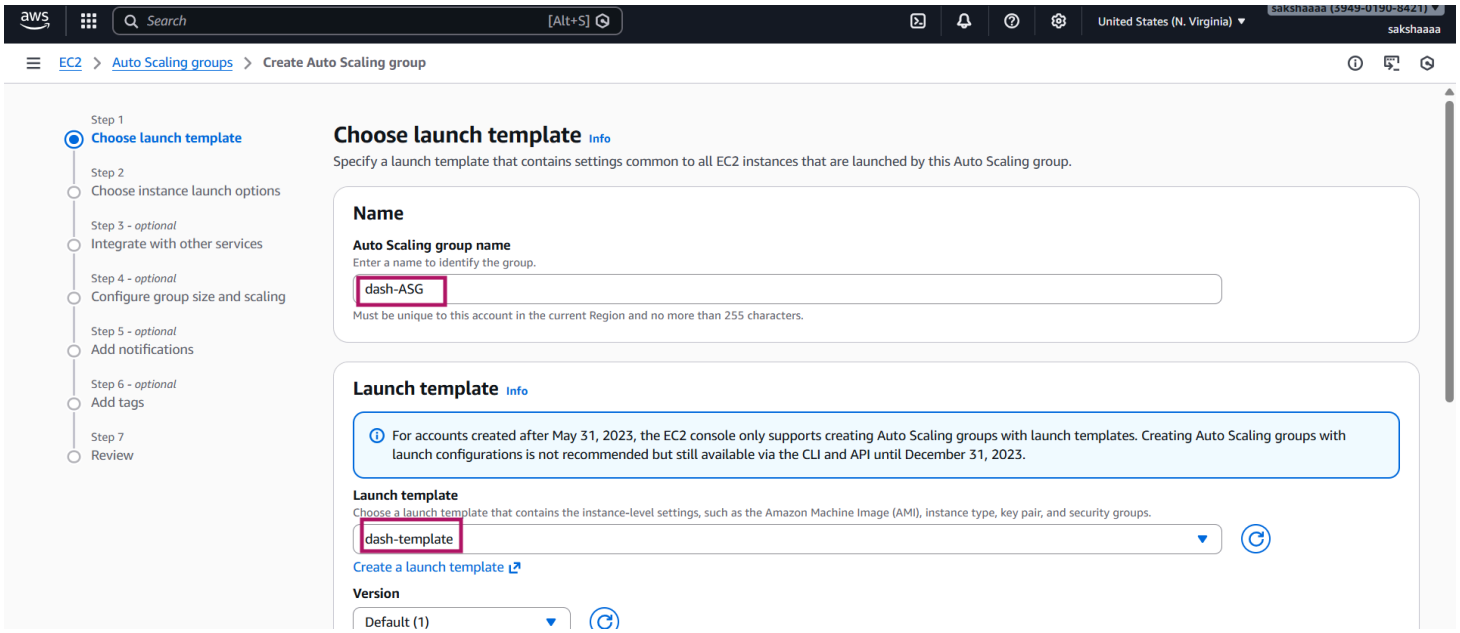
- Select custom AMI
- Choose instance type (t2.micro)
- Attach security group and key pair
- Create launch template



Step 4 : Create Auto Scaling Group

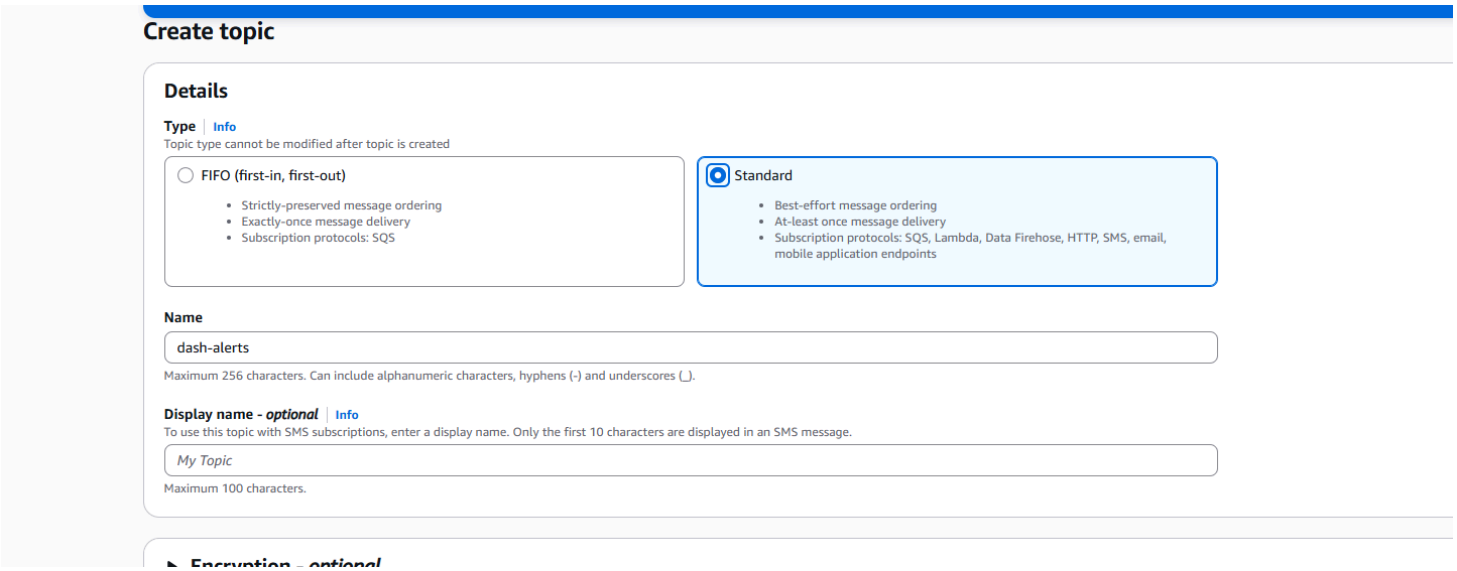
Configuration:

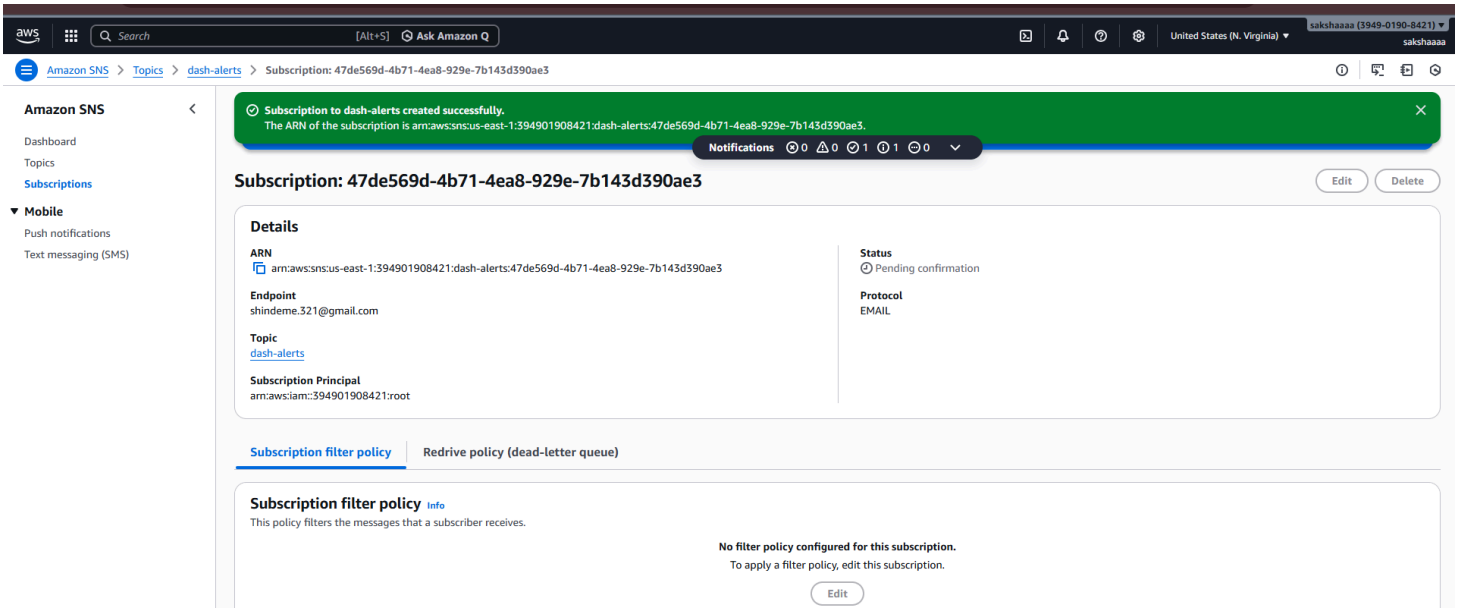
- Minimum capacity: 1
- Desired capacity: 1
- Maximum capacity: 3
- Select multiple subnets
- Enable health checks



Step 5 : Configure SNS Email Alerts

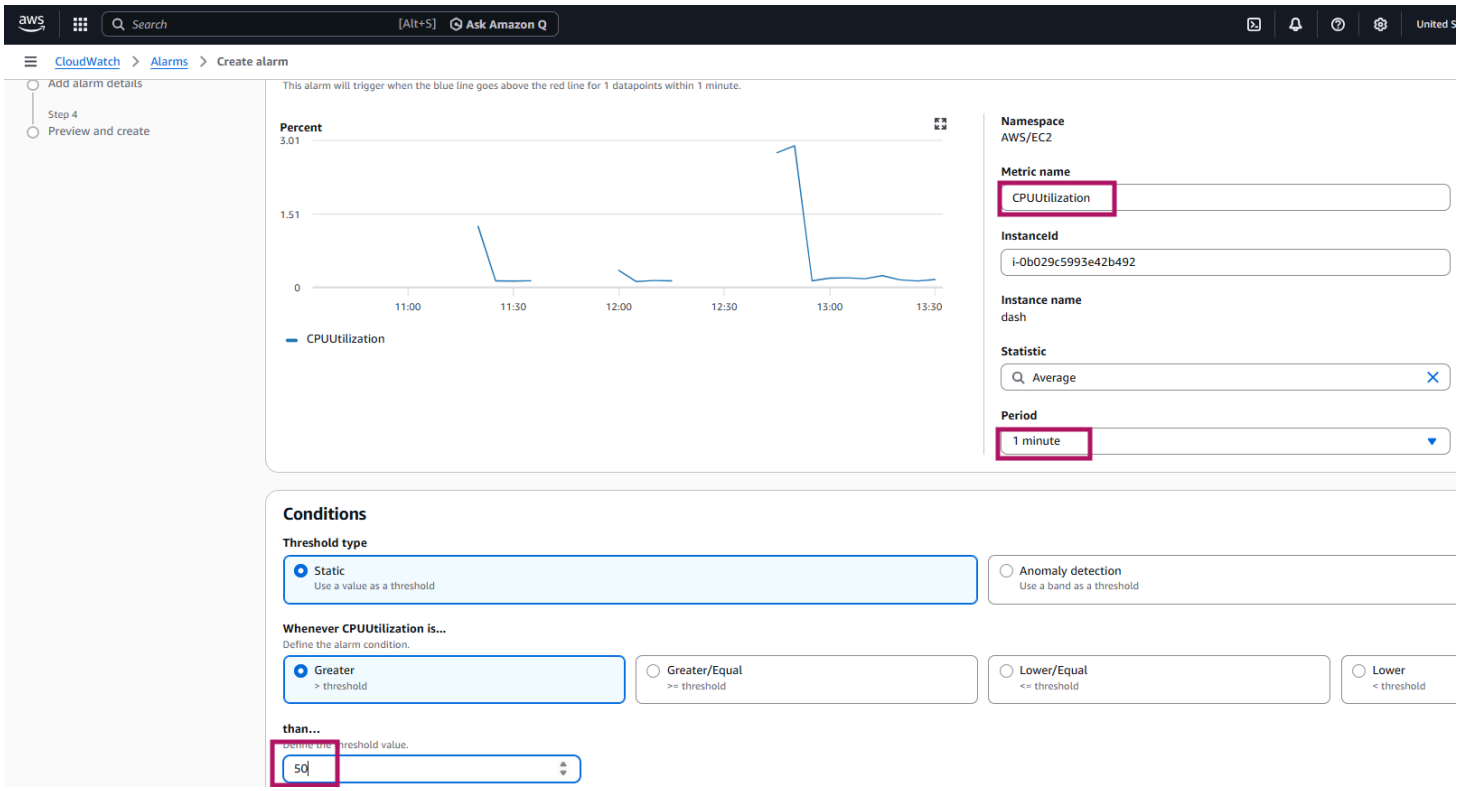
- Create SNS topic
- Create email subscription
- Confirm subscription from email





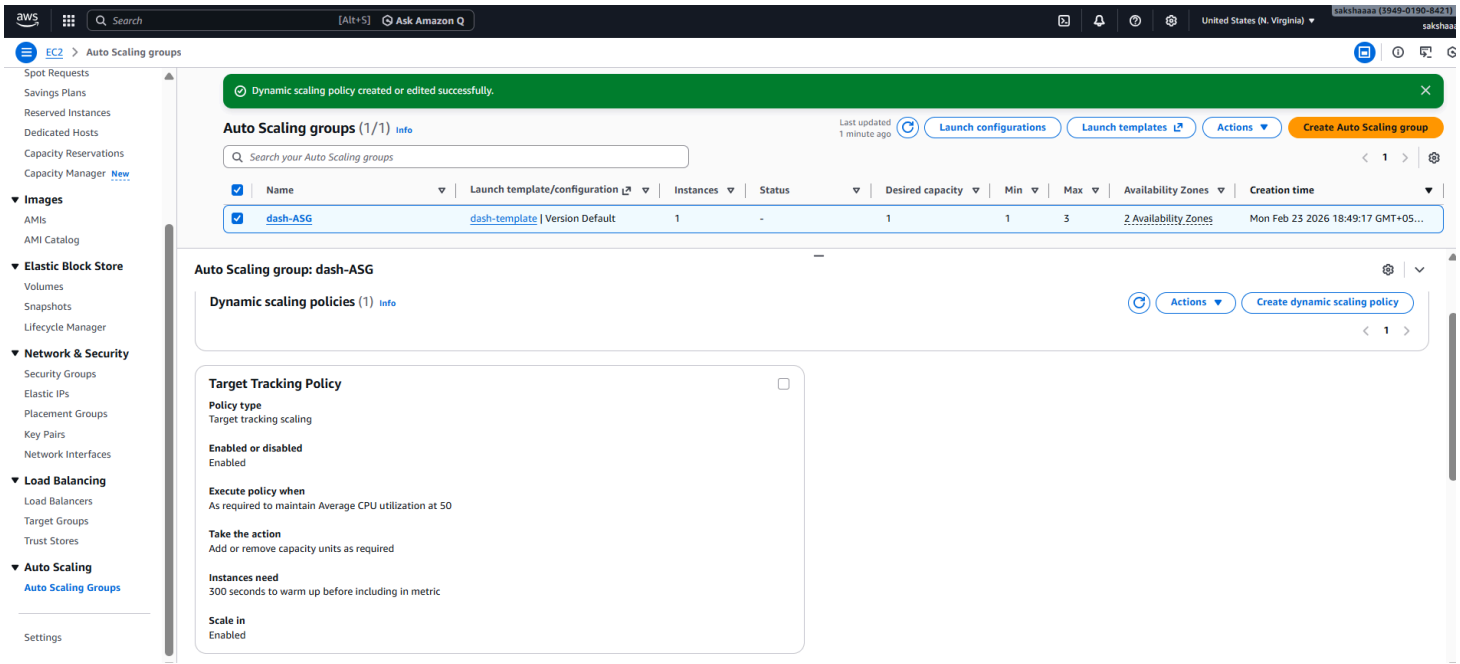
Step 6 : CloudWatch Alarm Configuration

- Monitor EC2 CPUUtilization
- Threshold: > 50%
- Attach SNS topic for notifications



Step 7 : Create Dynamic Scaling Policy

- Policy type: Target tracking
- Metric: Average CPU utilization
- Target value: 50%
- Instance warmup: 300 seconds



Step 8 : Testing and Validation

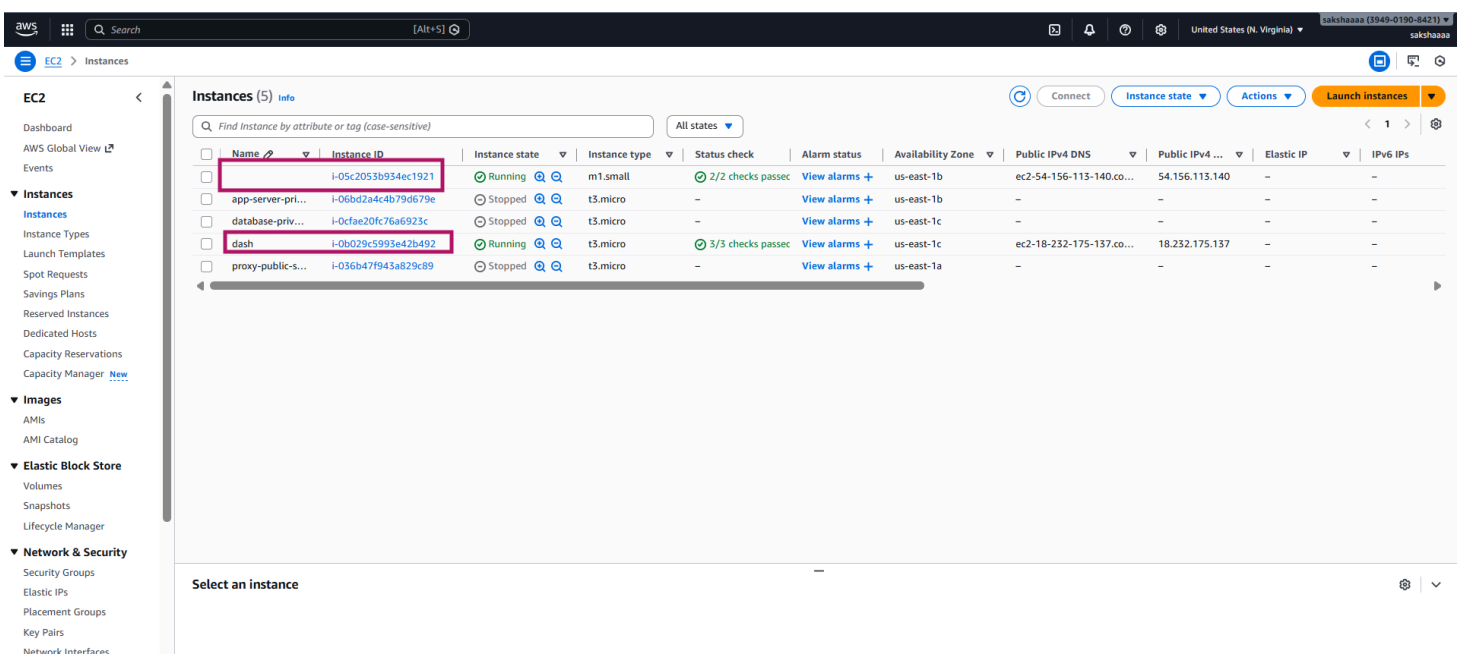
- Generate artificial CPU load using stress tool

Test Command:

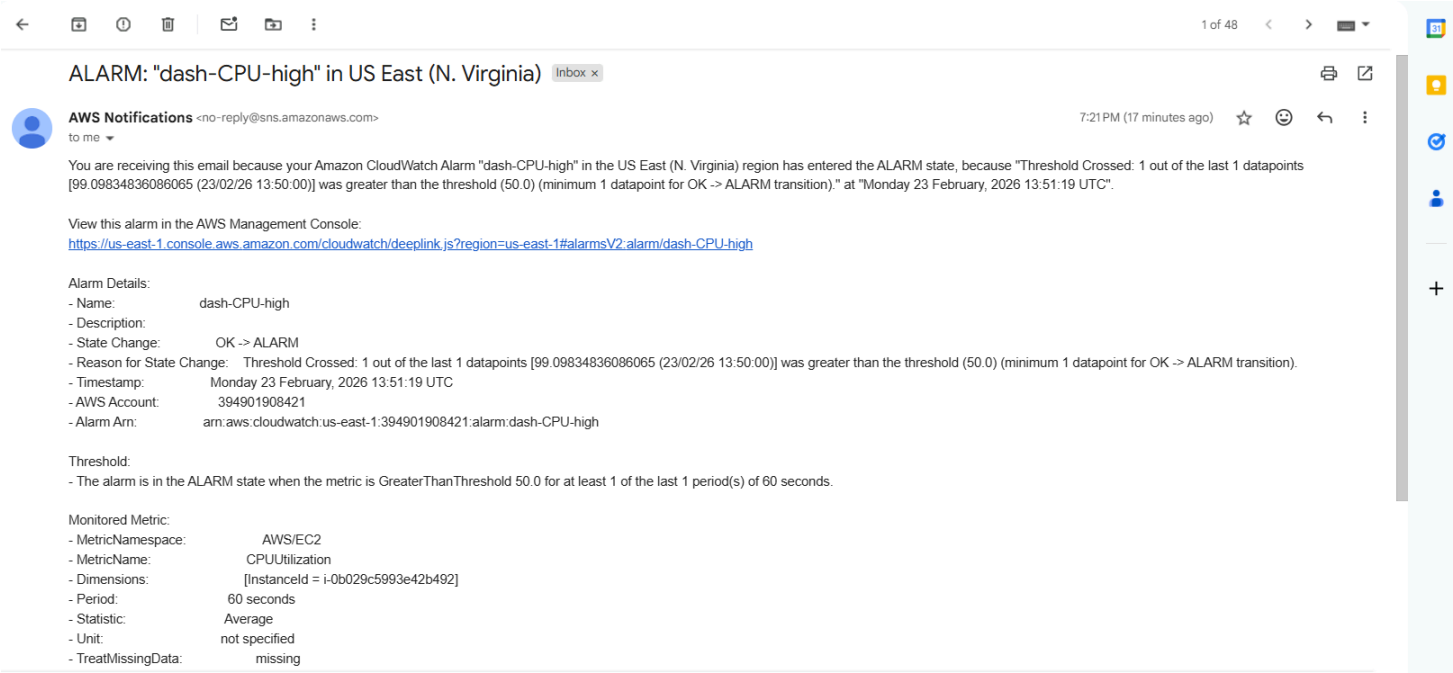
- sudo yum install stress -y
- tress --cpu 2 --timeout 300

```
[ec2-user@ip-172-31-24-61 ~]$ sudo yum install stress -y
Last metadata expiration check: 1:00:04 ago on Mon Feb 23 12:48:58 2026.
Dependencies resolved.
================================================================================================================================
 Package                        Architecture                   Version                                    Repository                          Size
================================================================================================================================
Installing:
 stress                         x86_64                         1.0.7-2.amzn2023.0.1                       amazonlinux                         34 k

Transaction Summary
================================================================================================================================
Install  1 Package

Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm                                                                   755 kB/s |  34 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                    398 kB/s |  34 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                            1/1
  Installing       : stress-1.0.7-2.amzn2023.0.1.x86_64                                                                         1/1
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64                                                                         1/1
  Verifying        : stress-1.0.7-2.amzn2023.0.1.x86_64                                                                         1/1

Installed:
  stress-1.0.7-2.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-24-61 ~]$ stress --cpu 2 --timeout 300
stress: info: [27310] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
client loop: send disconnect: Connection reset
```

- Observe automatic EC2 launch



- Verify email notification
- Confirm identical application output



ALARM: "dash-CPU-high" in US East (N. Virginia) Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 7:21 PM (17 minutes ago)
to me

You are receiving this email because your Amazon CloudWatch Alarm "dash-CPU-high" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [99.09834836086065 (23/02/26 13:50:00)] was greater than the threshold (50.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Monday 23 February, 2026 13:51:19 UTC".

View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/dash-CPU-high

Alarm Details:
- Name: dash-CPU-high
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [99.09834836086065 (23/02/26 13:50:00)] was greater than the threshold (50.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Monday 23 February, 2026 13:51:19 UTC
- AWS Account: 394901908421
- Alarm Arn: arn:aws:cloudwatch:us-east-1:394901908421:alarm:dash-CPU-high

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 50.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:
- MetricNamespace: AWS/EC2
- MetricName: CPUUtilization
- Dimensions: [InstanceId = i-0b029c5993e42b492]
- Period: 60 seconds
- Statistic: Average
- Unit: not specified
- TreatMissingData: missing

Summary :

SmartScale is an AWS-based automation project that demonstrates intelligent EC2 auto scaling driven by real-time CPU monitoring. An EC2 instance running Nginx is continuously monitored using Amazon CloudWatch, and when CPU utilization exceeds the defined threshold, an Auto Scaling Group automatically launches a new instance using a custom AMI to maintain performance and availability. The system also integrates Amazon SNS to send real-time email alerts during scaling events. This project showcases a production-style, self-healing cloud architecture that improves reliability, scalability, and operational efficiency.